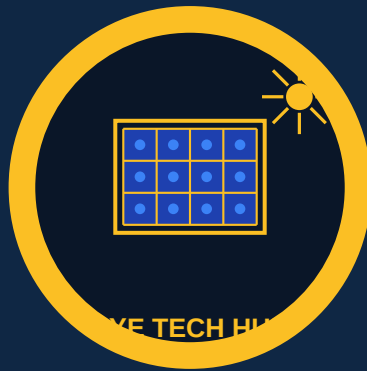




Solar Panels In Depth

I-V Curve · MPP · Temperature · Shading · Wiring

• SOLAR ENERGY • LESSON 04

The I-V Curve — A Panel's DNA

Every solar panel has a characteristic **Current-Voltage (I-V) curve** that describes how much current it delivers at every possible operating voltage. This curve is the most important piece of information on any panel datasheet. Everything else — energy yield, string sizing, inverter matching, temperature derating — follows from understanding it.

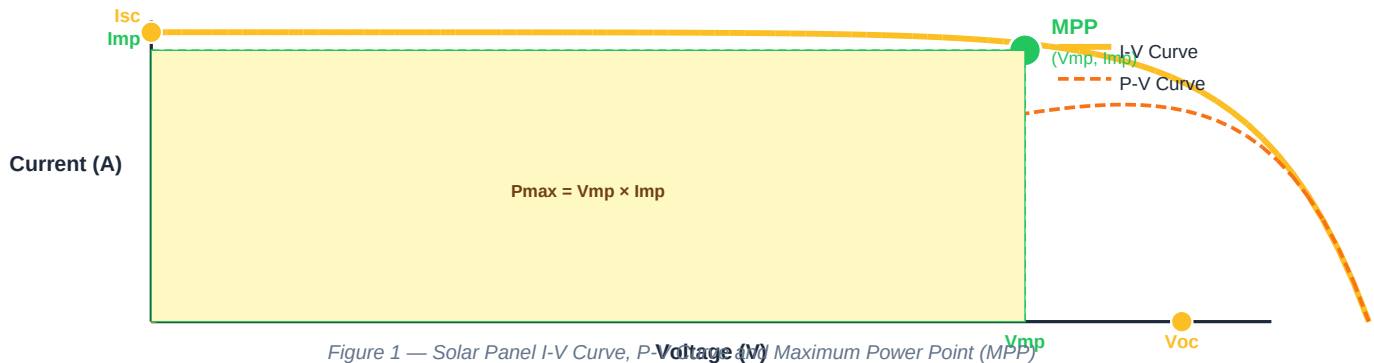


Figure 1 — Solar Panel I-V Curve, P-V Curve, and Maximum Power Point (MPP)

Point / Region	Symbol	Definition	Typical Value (400 W panel)
Short-Circuit Current	Isc	Current when output terminals are shorted (V=0). Maximum current the panel can produce.	~10.5 A
Open-Circuit Voltage	Voc	Voltage when no current flows (open circuit). Maximum voltage across terminals.	~41.5 V
Maximum Power Current	Imp (or Impp)	Current at the Maximum Power Point — the operating point that yields peak power.	~9.9 A
Maximum Power Voltage	Vmp (or Vmpp)	Voltage at the Maximum Power Point.	~40.4 V
Maximum / Peak Power	Pmax (Wp)	$P_{max} = V_{mp} \times I_{mp}$. Rated at STC (1 000 W/m ² , 25°C, AM1.5).	400 Wp
Fill Factor	FF	$FF = P_{max} / (V_{oc} \times I_{sc})$. Measures how "square" the I-V curve is. Higher = better quality.	~0.80 (good quality)
Knee Region	—	The rapid drop-off zone where voltage rises sharply but current falls — avoid operating here.	Near Voc; bypass diodes protect this region

MPPT explained: The inverter's MPPT tracker continuously adjusts the operating voltage of the string to stay exactly at V_{mp} — the peak of the P-V curve. As irradiance, temperature, and shading change throughout the day, V_{mp} shifts and the MPPT follows it in real time.

Temperature Effects and the Panel Datasheet

Solar panels are rated at **STC — Standard Test Conditions**: 1 000 W/m² irradiance, 25°C cell temperature, AM1.5 spectrum. In the real world, panel temperatures regularly reach 50–70°C on a hot summer day. This causes a significant drop in output that every system designer must account for.

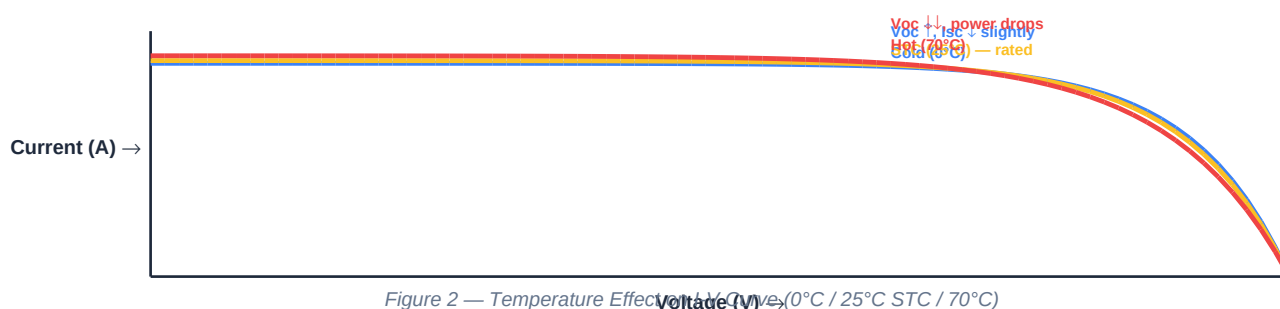


Figure 2 — Temperature Effect on I-V Curve (0°C / 25°C STC / 70°C)

Temperature Coefficient	Typical Value	Effect at 65°C Cell Temp (vs 25°C STC)
Power (Pmax) — γ	–0.35 to –0.45 %/°C	40°C above STC → –14% to –18% power loss A 400 W panel produces only 332–344 W
Voltage (Voc) — β	–0.27 to –0.36 %/°C	Voc drops → affects string voltage for inverter MPPT range
Current (Isc) — α	+0.04 to +0.06 %/°C	Isc rises slightly — small benefit; does not offset voltage drop
NOCT (Nominal Operating Cell Temp)	42–48°C (varies by panel)	Cell temp = Ambient + (NOCT – 20°C) × (G / 800) Use for realistic daily yield estimates

How to Read a Solar Panel Datasheet

Datasheet Field	What to Check	Red Flags
STC Power (Wp) and tolerance	Panel wattage; accept +3%/–0% tolerance (positive tolerance)	Negative tolerance (–3%) means delivered power could be below rated
Efficiency (%)	Power per unit area. >21% = premium mono; 16–18% = entry-level	Very cheap panels with claimed >23% efficiency — verify with test cert.
Voc and Vmp	Use Voc × 1.15 safety factor for inverter max DC voltage check	String Voc must never exceed inverter max input voltage (e.g. 1 000 V)
Isc and Imp	Size fuses and DC cable to 1.25 × Isc per string	If combiner box fuses are undersized, short circuit can cause fire
Temperature coefficients	Lower (less negative) γ = better in hot climates	Cheap panels often omit NOCT — assume 48°C if not stated
Degradation / Warranty	Linear power warranty: ≤0.5%/yr degradation to 80% at year 25	Step-function warranty (e.g. 90% at yr 10, 80% at yr 25) is industry minimum
IEC/UL Certifications	IEC 61215 (performance) + IEC 61730 (safety) are mandatory minimum	No third-party certification = significant quality and warranty risk

Bypass diodes: Each panel contains 3 bypass diodes (one per cell group). When a cell is shaded, its diode activates and routes current around the shaded group — preventing the "Christmas lights effect" where one shaded cell kills the whole string. The shaded section loses ~33% of panel power, not 100%.

Wiring Panels: Series, Parallel and Shading

How you connect panels determines the voltage and current delivered to the inverter. Getting this right is critical for inverter compatibility, energy yield, and safety. Two rules always hold: **series adds voltage, parallel adds current**. Power is always $V \times I$.

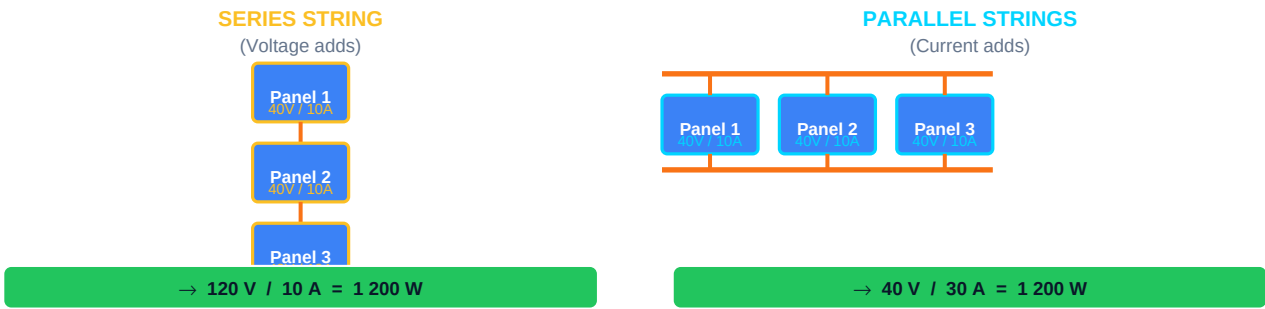


Figure 3 — Series Wiring Adds Voltage; Parallel Wiring Adds Current

Wiring	Voltage	Current	Power	When to Use
3 panels in series (40V / 10A each)	$3 \times 40 = 120\text{ V}$	10 A	1 200 W	Higher voltage strings reduce cable losses; match inverter MPPT range
3 panels in parallel (40V / 10A each)	40 V	$3 \times 10 = 30\text{ A}$	1 200 W	Low-voltage battery systems; when strings must be short
2S $\times$ 2P (2 series $\times$ 2 parallel)	$2 \times 40 = 80\text{ V}$	$2 \times 10 = 20\text{ A}$	1 600 W	Balance voltage and current; used in combiner boxes with multiple strings

Shading Losses — The Biggest Silent Killer

Shading Scenario	Loss Without Bypass Diodes	Loss With Bypass Diodes	Solution
1 cell fully shaded in a 60-cell panel	Up to 100% of string power lost	~33% of panel power (1 of 3 cell groups bypassed)	Bypass diodes (standard on all modern panels)
1 panel shaded in a 10-panel string (no optimizer)	50–80% string power loss	10–30% string power loss (bypass diodes active)	Microinverters or DC power optimisers per panel
Partial shade from nearby building (AM/PM)	Year-round loss of 15–25% energy yield	Reduced with optimisers	Reorient array; use east-west split; shade analysis in PVsyst
Bird droppings / soiling on 1 panel	Same as 1 panel shaded	Same as 1 panel shaded	Regular cleaning; tilt $\geq 10^\circ$ for self-cleaning by rain

Panel Selection Decision Guide

Priority	Best Technology	Why
Max output, limited roof space	Monocrystalline 420–440 Wp (half-cut, multi-busbar)	Highest efficiency (21–23%); best W/m ² ; lowest BOS cost per kWp
Hot climate performance	Mono with low temperature coefficient ($\gamma < -0.35\%/^\circ\text{C}$)	Less power lost on hot days; bifacial adds ground-reflected gain
Shade-affected roof	Microinverters or DC optimisers + any good mono panel	Panel-level MPPT eliminates string mismatch; worth the extra cost
Large ground-mount (cost priority)	Bifacial mono, large format (480–580 Wp), tracker optional	Bifacial adds 5–15% yield; large format lowers installation labour

Budget residential	Standard 400–410 Wp mono + string inverter	Best value; reliable; 10+ year proven track record; widely available
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What comes next — SOLAR-005: Solar Batteries — lead-acid vs LFP, state of charge, depth of discharge, battery sizing calculations, and how to specify the right battery bank for your system.